

Conglomerate Mergers: Effects on Growth and Competition

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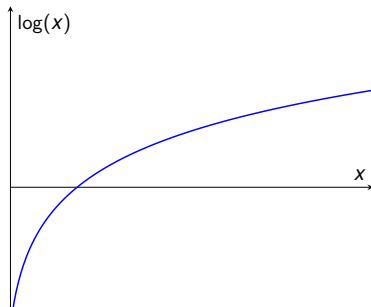
Motivation



- A conglomerate is a combination of business units operating in entirely different industries
- Usually they span multiple countries (multinational)
- There are more than 1,000 conglomerates worldwide
- Highly dynamic process of merging, acquiring and splitting
- Many of the brands you use daily probably belong to a conglomerate

Growth Optimization: A Primer

- “When tomorrow depends on today”
- Losses hurt exponentially more
- Irreversibility of state transitions makes things even worse (Think Russian Roulette)
- Most economic processes are multiplicative
 - Capital accumulation
 - Trust, reputation
 - Legal cases
- Shifting the return distribution upwards, at the smallest cost



Key Questions

- 1 Conglomerate mergers are not well understood in economics. Why would firms in unrelated markets merge?
- 2 How can antitrust authorities, who are tasked with protecting consumers from harmful mergers, evaluate these mergers?
- 3 How will transaction costs (communication, transfer, trust, etc.) affect conglomerate structure?

Model and setup

Input: $N, M, C, \Delta, T, p, \alpha, \lambda$

for t **in** T **do**

foreach $m_j \in N \times M$ **do**

 Merge with m_j if $m_j \notin n_i$ and
 $m_j \notin c_{it}$ with p

end

foreach $c_k \in C$ **do**

$pool_k = \sum_{m_j \in c_k} \alpha \Delta_{it} - \lambda(c_k)$
 $\Delta_{it}^* = (1 - \alpha) \Delta_{it} + \frac{pool_k}{\#m_j \in c_k}$

end

foreach $m_j \in N \times M$ **do**

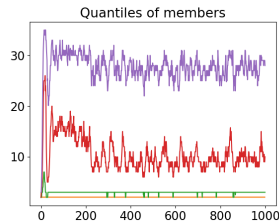
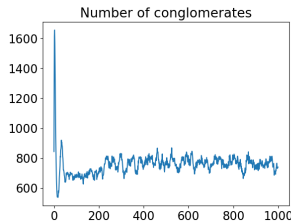
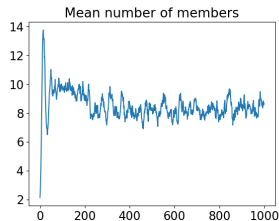
 Split if $s_{it} < s_{it-4}$

end

end

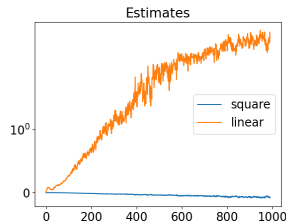
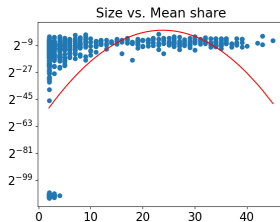
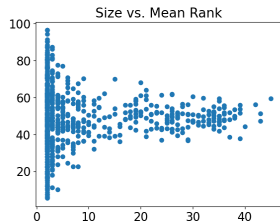
- $N = \#$ Markets (100)
- $M = \#$ Firms per Market (100)
- $C =$ Conglomerates
- $\Delta =$ Growth ($\sim \mathcal{N}(\mu_n, \sigma_n)$)
- $T =$ Total steps (1000)
- $p =$ Merge probability (0.02)
- $\alpha =$ Pooling rate (0.2)
- $\lambda =$ Transaction cost function
 - $\frac{\partial \lambda}{\partial \#m_j \in c_k} > 0$
 - $\frac{\partial^2 \lambda}{\partial^2 \#m_j \in c_k} > 0$

Merging Behavior



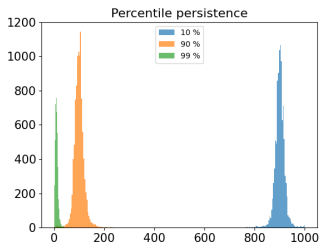
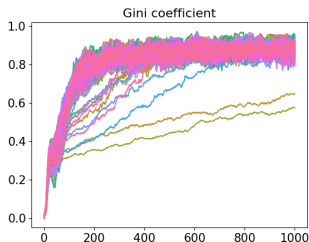
- Merging is a dominant strategy with more than 70 % of firms in a conglomerate (invariant to merge probability p)
- Stabilization in the number of conglomerates (convergence speed depends on transaction costs)
- Highly unequal distribution of conglomerate size, with the 90 % and 70 % quantiles being far apart from the 25 % and 10 % ones

Market Concentration (1/3) - Size Matters



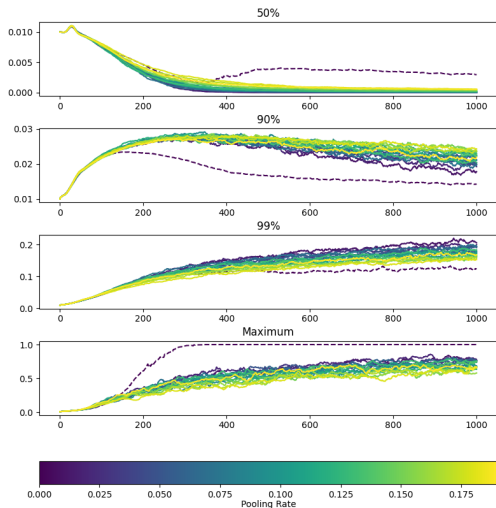
- Increasing the conglomerate size decreases the chance to end up in the lower quantiles (ranks) and to have lower market share ($\frac{s_{it}}{\sum_i s_{it}}$)
- Increasing the size above a certain threshold will not increase the success anymore (inverse U-shape)
- In the progression of the model the optimal size increases, but the costs eat up the gains from size at an increasing rate

Market Concentration (2/3) - Persistence & Mobility



- Almost all markets tend to strong, but not perfect inequality
- Many market participants have almost no market share
- High inequality is driven by low mobility across the ranks
 - Maximum quantiles are populated by a few enduring firms
 - Low quantiles are made up by a varying population
 - High end (90 %) quantiles are much more attainable

Market Concentration (3/3) - The Impact of Pooling



- No-pooling baseline case (dotted line) yields the known result of high skew
- Increasing the pooling rate marginally (to 1 %) from the baseline already lowers market concentration drastically
- Increasing the sharing rate lowers market concentration gradually

Key Findings

- ① Including dynamic frameworks allows for the explanation of many phenomena of economic exchange
- ② Increasing firm cooperation through conglomerate formation does not necessarily harm competition
- ③ Transaction costs play a crucial role in determining the size of a conglomerate

Model Extension: Endogenous Pricing

Input: $N, M, C, \Delta, T, p, \alpha, \lambda$

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foreach $m_j \in N \times M$ **do**

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end

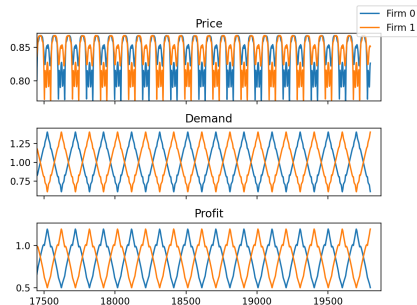
Model Extension: Endogenous Pricing

```
⋮  
foreach  $m \in M$  do  
  |  $p_{mt} = \pi(\cdot)$   
end  
foreach  $c_k \in C$  do  
  |  $pool_k = \sum_{m_i \in c_k} \alpha \Pi_{it}(\cdot) - \lambda(c_k)$   
  |  $\Delta_{it}^* = (1 - \alpha)\Delta_{it} + \frac{pool_k}{\#m_i \in c_k}$   
end  
foreach  $m \in M$  do  
  |  $s_{mt}(\Pi)$   
end  
⋮
```

- Allow firms to set prices following their policy $\pi(\cdot)$
- Fluctuations in a company's growth through stochastic costs c_{it}
- Market share as the cumulative profits
 $\Pi_{it}(\mathbf{P}_t, s_{it}) = D(\mathbf{P}_t, s_{it}) * p_{it} - c_{it}$ with
 $s_{it}(\Pi) = \frac{\sum_{\tau=1}^t \Pi_{i\tau}}{\sum_{\tau=1}^t \sum_{j:j \neq i} \Pi_{j\tau}}$

Model Extension: Pricing Decision

- Demand function of the form
$$D(\mathbf{P}_t, s_t) = 1 - (1 - s_{it})p_{it} + 0.5 \max_{j:j \neq i} p_{jt}$$
- Learn the optimal pricing policy $\pi(\cdot)$ using a Proximal Policy Optimization reinforcement learning algorithm
- Information set of the firm
$$\{p_{it-1}, s_{it-1}, \max_{j:j \neq i} p_{jt-1}, \max_{j:j \neq i} p_{jt-2}\}$$
- Goal of the firm is to optimize the profit
$$\Pi_{it}(\mathbf{P}_t, s_{it}) = D(\mathbf{P}_t, s_{it}) * p_{it}$$



Questions to be addressed

- More dynamic splitting decisions depending on a change of ranks or market share
- Include jump processes
- Investigate whether certain industrial configurations have different underlying processes (chemicals, electronics, consumer goods, etc.)
- Check how conglomerate performance is linked to the underlying growth process
- Find stronger transaction cost modelling
- Profit pooling vs. Cost pooling

Reach Out!

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